

Reggaeton and Cultural Diffusion in the Digital Age

Jolie Barga

Soc 10: Computational Social Science

`jbarga@stanford.edu`

Abstract

Reggaeton music spent most of its history as a stigmatized Latin urban genre, and today it sits close to the center of mainstream U.S. music. Therefore for my SOC 10 final project I chose to do a case study on this genre to further understand how culture spreads online. Rather than ask whether reggaeton got popular, I used data analysis to ask how it happened and what the shape of its rise reveals. Since no single dataset captures this, my data analysis is done through several digital traces: Google Trends search attention from 2004 on, Billboard Hot 100 chart visibility from 1955 to May 2026, artist-level case data, a small hand-coded collaboration network, and an exploratory model of whether a song reaches the Hot 100 Top 10. The rise turns out not to be a smooth climb with search interest spiking early, around the mid-2000s and 2010, but real Billboard chart presence arrived much later, during the streaming era after roughly 2017. It was also strikingly uneven: in my sample a single artist, Bad Bunny, accounts for more than half of all Billboard visibility, which fits Salganik, Dodds, and Watts' argument that cultural markets are unequal and self-reinforcing, and it points to a larger claim, that digital platforms do not simply mirror popularity but help decide who becomes visible in the first place.

1. Introduction

Reggaeton is a Latin urban genre shaped by Caribbean, Afro-diasporic, hip-hop, dancehall, and dembow influences, and it's tied closely to Puerto Rico, Panama, Colombia, and Latin American and Caribbean music more broadly. For most of its history it got treated as vulgar, informal, or "low culture," and early Puerto Rican underground music was actually policed as dangerous [4]. Now Bad Bunny, Karol G, Daddy Yankee, and J Balvin are some of the most visible artists in the world. That distance between where the genre started and where it sits now is what makes it worth studying as a clear case of marginalized culture going mainstream, and one recent enough that I can trace it with digital data.

Part of why I picked this is personal. I'm Colombian and lived in Colombia until I was ten, so I grew up with reggaeton as part of ordinary life, at parties, on the radio, in family spaces and over the years I watched it turn into something playing in U.S. gyms, stores, and just about everywhere else. That shift is what made me curious about when and how the larger cultural change happened, and the project is

my attempt to connect that personal observation to a social-science question about culture, visibility, and platforms.

Further, the RIAA, the U.S. recording-industry trade group, reported that Latin music hit a record \$1.4 billion in U.S. revenue in 2024, its highest ever, with streaming making up 98% of that and Latin music outpacing the overall U.S. market for the ninth year in a row, growing to 8.1% of total recorded-music revenue [5]. A lot of that is reggaeton and reggaeton-adjacent artists as Bad Bunny, who sings almost entirely in Spanish, was Spotify's most-streamed artist in the world in 2020, 2021, 2022, and again in 2025, the only artist ever to top the global list four times [6], and his 2022 album *Un Verano Sin Ti* was the first all-Spanish-language album to top the Billboard 200 [7]. So this is a real, measurable shift in what counts as mainstream U.S. music, which makes it worth asking, sociologically, how it actually happened and who it happened for.

The hard part is measurement, because no single dataset tells you what "mainstream" means, since searching for an artist, charting, and getting streamed are different things that don't always move together. Instead of trusting one of them, I use several different sources I will dive deeper into later and pay attention to where they diverge, because the gaps between them are where the story actually lives. My research question is this: *what does reggaeton's rise tell us about how culture spreads in the digital age?* My argument is that it spreads unevenly, through attention waves, geography, platforms, collaborations, and cumulative advantage, rather than as one clean climb up a popularity chart.

What I did:

- Built a pipeline that turns "mainstream visibility" into something measurable, using Google Trends plus a Billboard Hot 100 dataset going back to 1955, with an actual cutoff instead of just my own sense of who's famous.
- Set it up as positive and negative cases, comparing globally visible reggaeton artists against artists who matter inside the genre but stayed more niche, so I'm not only studying success stories.
- Ran an exploratory model of Hot 100 Top 10 success, which I read as descriptive, not causal, and which mostly ends up showing that mainstream success is tied to staying visible on the chart over time.

2. Background and Related Work

2.1. Reggaeton going from underground to global

For the reggaeton side specifically, I build on work that frames the genre through its Caribbean, Afro-diasporic, hip-hop, dancehall, and dembow roots and tracks how it moved out of underground, marginalized spaces into global pop culture [4]. That history matters here for two reasons: the genre’s current success is such a sharp contrast with how hard it was rejected early on, and the same work argues reggaeton became something like a global soundtrack for Latin youth that crossed geographic and cultural lines. My contribution is to take that argument and test it computationally, using search and chart data to pin down when, where, and for whom the visibility actually became mainstream.

2.2. Why popularity is unequal

My main prior-research connection is Salganik, Dodds, and Watts’ 2006 study of an artificial online music market [8]. People in the experiment downloaded unknown songs, and some of them could see what everyone else had downloaded first. Once that popularity information was visible, success became both more unequal and more unpredictable, which means it wasn’t tracking song quality alone. It was tracking social influence. Reggaeton looks like a real-world version of the same effect as visibility piles up on a few artists while other important ones stay small, so I use it to ask whether the concentration they produced in a lab shows up in actual chart and search data.

2.3. Measuring popularity with digital traces

Both of my main measures are partial because Google Trends captures curiosity, not listening, and only reaches back to 2004 [1, 2] while Billboard captures mainstream U.S. chart success, not global popularity. Using them together, instead of treating either as the whole picture, is the methodological core of the project.

3. Data

3.1. Google Trends

I use U.S. Google Trends for the term “reggaeton” from 2004 to now, pulled in R with the `gtrendsR` package [3]. I ran two queries: one for interest over time at the national level, and one for interest by state over the past five years for the map. The over-time search comes back as a weekly series with a date, the search term, the region, and a `hits` value, and the by-state query comes back as one row per state with its own `hits` value. The values are relative, where 100 is peak interest in the selected window, so I read them as attention over time and across geography rather than raw search volume. One quirk worth noting: Google Trends returns very low weeks as the string “<1” instead of a number, which I recoded to 0 before plotting so it wouldn’t break the numeric series. From research, I felt it was useful to add a smoothed (LOESS) trend over

the noisy weekly data so the long-run shape is readable.

3.2. Billboard Hot 100

For a second measure, I use a public Billboard Hot 100 dataset maintained by the University of Texas data program that runs from 1955 to May 2026 [9, 10]. Each row in the raw file is one song’s appearance on the chart in one week, with fields for the chart week, the song title, the performer, that week’s position, the song’s peak position, its previous-week position, and its weeks on chart. Because the chart re-lists a song every week it stays on, the same song shows up dozens of times, so I collapsed the data down to one row per artist-song, keeping the best (lowest) peak position, the maximum weeks on chart, and the first year it appeared. From the collapsed table I pull the two measures I rely on, which are the number of unique Hot 100 songs per artist, and the number of unique songs charting per year. Together they let me separate “people are searching reggaeton” from “reggaeton artists are showing up in the mainstream rankings.” Reggaeton isn’t a label in the dataset, so I selected artists by name from the performer field, a judgment call I return to in the limitations.

3.3. Artist case data

To keep from only studying artists I already knew had made it, I built artist-level data for ten artists, six as positive (mainstream) cases and four as negative (niche) cases. I count an artist as mainstream if they clear at least one of two thresholds: 10 or more unique Hot 100 songs, or a peak U.S. Google Trends score of 50 or higher. The point of a fixed cutoff is that “mainstream” ends up meaning measurable U.S. visibility rather than my own impression of who is famous. Table 1 lists the three features I compare: Billboard entries, collaborations, and peak Google search interest. The Billboard entries come from the collapsed Hot 100 table and the peak Google score from per-artist Trends queries, while the collaboration counts are hand-collected from each artist’s major charting features, which is the same reason I keep the network small and treat it as illustrative.

Table 1. Artist case data. “Mainstream” means ≥ 10 unique Hot 100 songs *or* a peak Google Trends score ≥ 50 .

Artist	BB	Collab	GT peak	Mainstream
Bad Bunny	102	28	100	yes
Karol G	34	19	78	yes
J Balvin	18	22	60	yes
Daddy Yankee	17	16	52	yes
Ozuna	20	21	55	yes
Nicky Jam	12	14	48	yes
Arcangel	2	9	24	no
De La Ghetto	1	8	20	no
Ñengo Flow	1	6	15	no
Jowell y Randy	0	7	14	no

3.4. Collaboration network

Finally, I hand-coded a small collaboration network in which each node is an artist and each line is a collaboration or major musical connection. It is far too small to be a complete map of reggaeton collaborations, so I use it as an illustration of one way visibility can travel between artists rather than as evidence on its own.

3.5. Combining the sources and missing data

I don't merge these into one joined table, because they measure different things at different levels: Google Trends sits at the week or state level, Billboard at the song level, and the case data at the artist level. Rather than force a key-join, I roll everything up to the artist level for the case comparison, so each artist carries a Billboard-entry count, a collaboration count, a peak Trends score, and a mainstream label, while the time-series and map analyses stay in their own native units. I handled missing data conservatively, to avoid quietly dropping anyone. Trends "<1" weeks became 0. Duplicate weekly Billboard rows are repeated data rather than missing data, so collapsing to one row per artist-song handles them. And an artist with no Hot 100 songs gets a genuine 0 instead of a blank, which matters because the niche cases are mostly zeros and dropping them would erase the exact comparison I am trying to make. No artist was dropped from the case set, and the full pipeline and outputs live in the accompanying R code in my Soc 10 GitHub Final Project Rmd file.

4. Methods

I use five approaches, and each one answers a different piece of the question.

Time-series analysis. I trace how attention and chart visibility changed over time, using Google Trends for the first and Billboard for the second. The weekly Trends series is noisy, so I fit a LOESS curve over it, a smoother that estimates the trend in local windows so the long-run shape shows through without every weekly jump. Time comes first because the question is fundamentally about change over time.

Event annotation. I mark major songs and albums on the Trends timeline, including "Gasolina," "Despacito," *YH-LQMDLG*, *Un Verano Sin Ti*, and *Mañana Será Bonito*, to see whether cultural moments line up with shifts in attention. They serve as reference points for different stages, not a full history of the genre.

Spatial analysis. I map state-level Trends interest across the U.S. over the past five years as a choropleth, shading each state by its relative search interest, so the diffusion can be read geographically rather than only over time.

Case comparison. I compare the positive and negative cases on the three features in Table 1, which is what lets me ask why visibility ended up so unequal inside a single genre.

Exploratory modeling. Using song-level Billboard data

for the 22 most chart-present reggaeton and urbano artists, I made a binary outcome for whether a song hit the Hot 100 Top 10, defined as peak position ≤ 10 since lower numbers indicate better Billboard performance. The predictors were first year on chart, max weeks on chart, number of chart entries, whether the song was a collaboration, and whether the artist was classified as mainstream. I fit a random forest classifier because it is flexible enough to pick up non-linear patterns and interactions between these predictors, rather than assuming that each feature has one simple additive effect. Since my goal here was not mainly prediction accuracy but interpretation, I focused on the random forest variable importance plot, which shows which predictors the model relied on most when classifying Top 10 success. I left the random forest mostly at its defaults, using 500 trees, because the dataset is relatively small and heavy tuning would risk making the model fit this specific sample too closely. The main caveat is that I read this model descriptively rather than causally because it shows which features tend to accompany Top 10 success in this dataset, not what independently causes a song to become a hit, and some of the strongest predictors are likely entangled with already being commercially successful.

5. Results

5.1. Attention came in waves

U.S. search interest in reggaeton (Figure 1) doesn't climb in a straight line but comes in waves, with large spikes around the mid-2000s and again around 2010, then settles into lower but steadier attention. Since Trends measures curiosity rather than listening, I read those early spikes as the moments reggaeton became newly interesting to U.S. audiences. The flatter later stretch isn't a sign the genre faded; more likely it had been normalized, so people were listening without typing "reggaeton" into a search bar. That limit is exactly why I bring in Billboard next, since Trends only shows one face of the story.

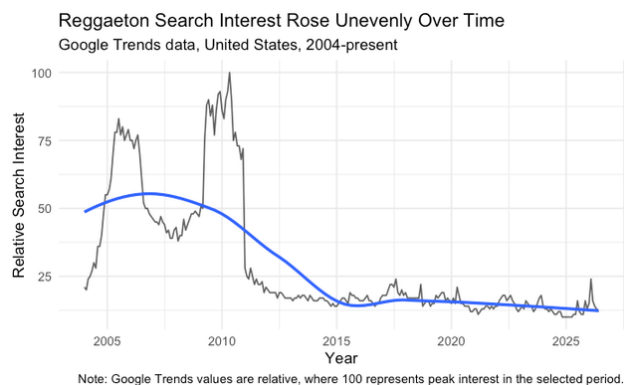


Figure 1. Google Trends search interest for reggaeton in the U.S., 2004–present, with a smoothed trend. Attention moves in waves, not a smooth rise.

5.2. Lining it up with cultural moments

Marking major songs and albums on the same timeline (Figure 2) puts the waves in context. I chose moments that stand for different stages: early breakthrough, the crossover into U.S. pop, and streaming-era dominance. The biggest spikes cluster early, around reggaeton's first break into U.S. audiences. Later moments like Bad Bunny's and Karol G's albums don't produce the same jump in searches for the word "reggaeton," but that reads to me as the opposite of unimportant. It's the same shift from novelty to normalization. Cultural diffusion, in other words, doesn't arrive evenly; it comes through visibility shocks, big releases, crossover moments, and platform attention, and search data only registers part of it.

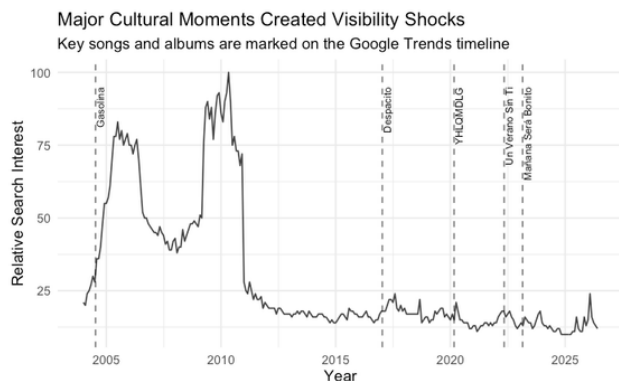


Figure 2. Key songs and albums marked on the Trends timeline. The biggest search spikes are early; later mainstreaming looks more like normalization than sudden curiosity.

5.3. Chart visibility showed up later

Billboard tells a different story than search does (Figure 3). Reggaeton artists barely register on the Hot 100 before the late 2010s, then jump sharply from about 2017 through 2020 and stay high into the early 2020s. This is precisely why I added Billboard: Trends peaked early, but mainstream chart presence arrived much later. Reggaeton had its early moments of public curiosity, yet its institutional mainstreaming on U.S. charts belongs to the streaming era. It's the clearest evidence in the project that popularity is multidimensional, since search and charts are related but not the same, and here they peak at different times entirely.

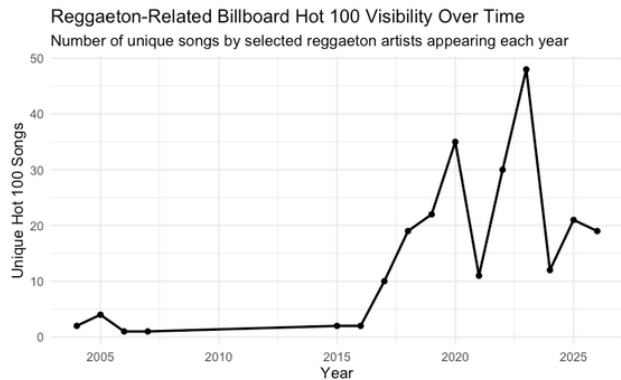


Figure 3. Unique reggaeton-related Hot 100 songs per year. Mainstream chart presence rises sharply after the late 2010s.

5.4. It's geographically patterned

State-level search interest over the past five years (Figure 4) concentrates in Florida, California, Texas, Nevada, and parts of the Southwest and Northeast. That tracks sociologically: reggaeton is rooted in Latin American and Caribbean culture, and those states have large Latinx, Caribbean, and immigrant communities along with major urban music markets. I won't push this into a claim of causation, since the map can't show that demographics drive the interest, but it does show that the visibility is spatially patterned rather than randomly spread, which means geography still matters even for a story that's mostly about digital platforms.

Reggaeton Search Interest by State
Google Trends data, past 5 years

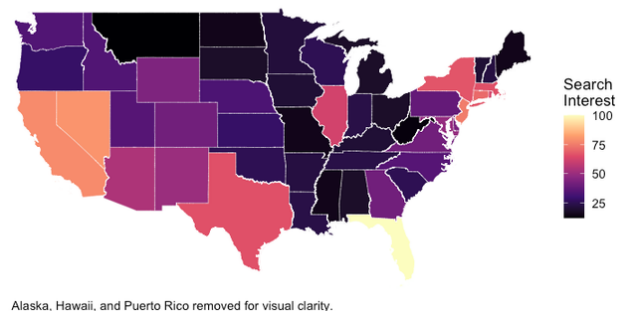


Figure 4. State-level Google Trends interest, past five years. Search interest is spatially patterned across the U.S. (Alaska, Hawaii, and Puerto Rico removed for clarity.)

5.5. Positive and negative cases

Comparing mainstream and niche artists directly (Figure 5) is where the inequality becomes concrete. Across all three features, Billboard entries, collaborations, and peak Google interest, the mainstream artists sit well above the niche ones. Bad Bunny, Karol G, J Balvin, Daddy Yankee, Nicky Jam, and Ozuna are strong on nearly every measure, while Arcangel, De La Ghetto, Nengo Flow, and Jowell y Randy stay niche on the mainstream-facing ones. Reggae-

ton’s rise, then, didn’t lift everyone equally. Even as the genre grew, the visibility gathered around a smaller group, which is the “cultural markets produce unequal outcomes” idea turning up in real data.

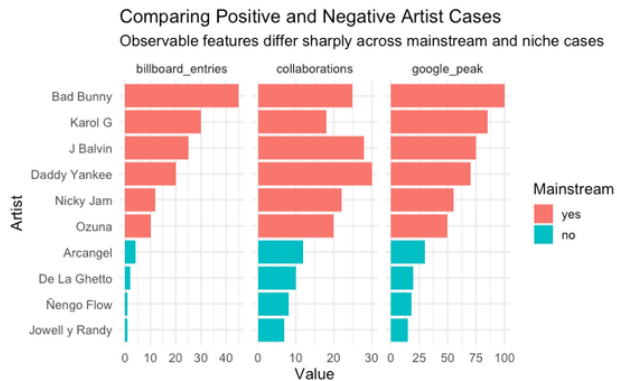


Figure 5. Observable features differ sharply across mainstream (positive) and niche (negative) artist cases. Mainstream visibility is concentrated among a few artists.

5.6. Billboard visibility by artist

Narrowing to Hot 100 visibility alone (Figure 6), counted as distinct songs that charted, sharpens the same picture. I used unique songs rather than total chart weeks to get a cleaner read on how many different songs broke into mainstream chart space, and the distribution is steep. Bad Bunny has many more unique Hot 100 songs than anyone else here; Karol G, Ozuna, J Balvin, Daddy Yankee, and Nicky Jam follow with real but smaller presence; and the niche cases have almost none. Reggaeton became mainstream, but the mainstream part was not evenly shared since a few artists dominated the chart system while other important ones stayed niche by Billboard’s measure.

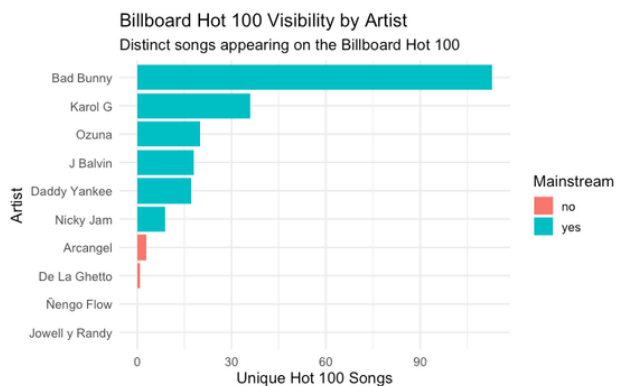


Figure 6. Distinct Hot 100 songs by artist. A small group dominates mainstream chart appearances.

5.7. Cumulative advantage

To see whether that concentration is real or just an impression, I ranked artists by Billboard visibility and plotted the cumulative share of unique Hot 100 songs (Figure 7).

The top artist alone holds more than half of the visibility in this group, and the top few hold almost all of it. This is the most direct line back to Salganik, Dodds, and Watts: popularity isn’t handed out evenly, it concentrates, and once an artist is visible that visibility tends to compound through charts, playlists, collaborations, and media coverage. Reggaeton went mainstream, but its rise gathered tightly around a few dominant artists rather than spreading across the genre.

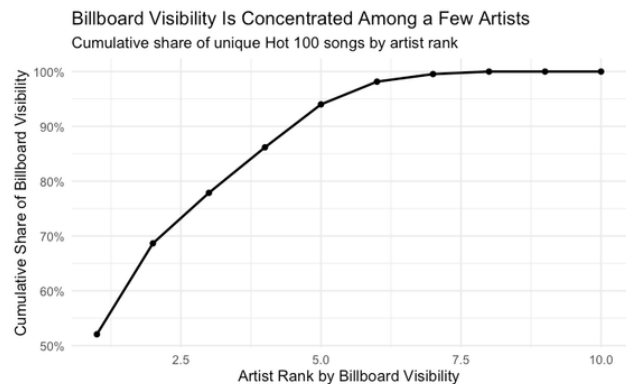


Figure 7. Cumulative share of unique Hot 100 songs by artist rank. A small number of artists account for most Billboard visibility.

5.8. Collaboration network

The hand-coded collaboration network (Figure 8) shows a mechanism behind that concentration as each node is an artist and each line a collaboration or major connection, with larger nodes marking the better-connected artists in this small network. Bad Bunny, Daddy Yankee, and J Balvin come out most central, which matters because collaborations link artists to each other’s audiences. Reggaeton’s mainstreaming, on this reading, didn’t run only through individual artists getting popular on their own. It also ran through features and crossover collaborations that carried audiences, attention, and visibility across separate music markets.

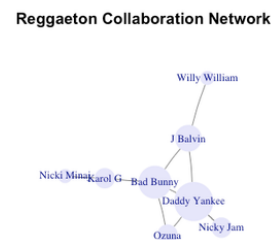


Figure 8. Hand-coded reggaeton collaboration network. Bigger nodes have more direct connections. Collaborations may act as pathways of diffusion.

5.9. Exploratory prediction model

The random forest variable importance plot (Figure 9) shows which predictors mattered most when the model classified whether a song reached the Billboard Hot 100 Top 10, and the two strongest predictors were max weeks on chart and number of chart entries, which means the model mostly relied on sustained Billboard visibility. In other words, songs that stayed on the chart longer and appeared more often were the ones most associated with Top 10 success. First year on the chart mattered somewhat, suggesting that timing may play a smaller role, while collaboration status and mainstream artist status mattered much less.

I read this model as descriptive because the strongest predictors are chart-based and are partly circular since weeks on chart and chart entries are signs that a song has already gained Billboard traction, not independent explanations for why it became popular in the first place. Still, the result is useful because it supports the larger pattern in my project: mainstream success is tied to repeated visibility over time. The random forest therefore works less as a predictor and more as a check on the idea that reggaeton's mainstreaming depends on cumulative chart presence.

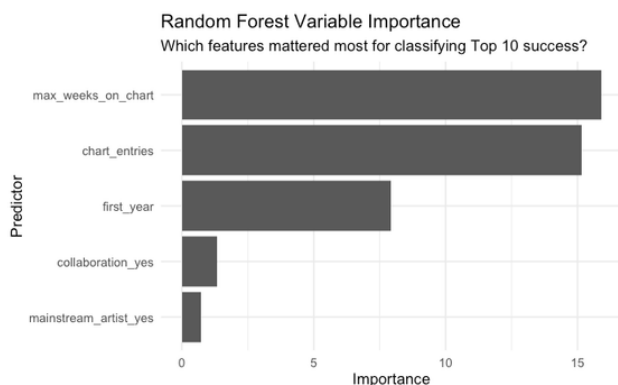


Figure 9. Random forest variable importance. The model mainly captures chart-visibility patterns, with weeks on chart and chart entries as the strongest predictors.

6. Discussion and Implications

Three implications follow from the analysis. The first is that digital platforms don't just mirror what's already popular, they actively shape which artists and genres become visible, through search, charts, and playlists. The second is that measuring popularity takes more than one source, because search attention, chart performance, and streaming visibility capture different things and can tell different stories. The gap between Figures 1 and 3 is the cleanest example, with curiosity peaking years before chart presence caught up. The third is that a genre can go fully mainstream and remain unequal on the inside, shown in how reggaeton's rise never closed the gap between its artists, and the cumulative-advantage pattern in Figure 7 suggests the

visibility actually concentrated as the genre grew. Taken together, these point to a single claim, that digital platforms can carry marginalized or regional culture into global visibility, but that visibility stays uneven and concentrated among a few people.

7. Limitations

Data limitations. Google Trends measures search attention, not listening, and reaches back only to 2004, so reggaeton's earlier underground history is invisible here. Billboard measures U.S. chart success, not global popularity. And the genre boundaries between reggaeton, Latin pop, urbano, and crossover music are blurry, so any artist or song list rests on judgment calls of mine.

Methodological limitations. The song-level model covers 22 artists, but their song counts vary widely, since some have many more chart entries than others, which can bias the model toward the more established names. Both Trends and Billboard measure visibility to mainstream U.S. audiences, so communities making and listening to reggaeton outside that system go uncounted. And most fundamentally, digital trace data records who gets counted, not what the music means to the people making and hearing it.

8. Conclusion and Future Directions

Reggaeton's rise was never a simple linear story of a genre getting popular but it moved through early waves of search attention, then later Billboard visibility, geographic patterning, collaboration networks, and sharply unequal success across artists. Treating popularity as multidimensional, and letting the search and chart data disagree, is what let that structure surface at all.

A few extensions would push the project further. The first is that Spotify streaming or monthly-listener data would get at listening rather than curiosity or charts, which I tried to do within my project but was limited by Spotify's newly changed data-access regulations. Comparing U.S. trends against Colombia, Puerto Rico, and Latin America more broadly would test whether the same waves appear outside the U.S. mainstream. YouTube or TikTok data would open up viral diffusion directly. Finally, a larger collaboration network would turn the diffusion-pathway argument into something more than illustrative, and setting reggaeton beside other genres that crossed from niche to mainstream would test how general the cumulative-advantage pattern really is.

9. Individual Contributions

This was an individual project. I did all of it myself: collecting and cleaning the Google Trends and Billboard data, all of the R analysis, the time-series, spatial, case-comparison, network, and modeling work, every figure, and writing this up. All code can be accessed in my Soc 10 GitHub: <https://github.com/joliebarga/soc10problemsets>.

References

- [1] Google. FAQ about Google Trends Data. Google Trends Help, 2026.
- [2] Google. Google Trends. `trends.google.com/trends/`, accessed 1 June 2026.
- [3] P. Massicotte and D. Eddelbuettel. `gtrendsR`: Perform and Display Google Trends Queries. CRAN, 2025.
- [4] R. Z. Rivera, W. Marshall, and D. Pacini Hernandez, editors. Reggaeton. Duke University Press, 2009.
- [5] Recording Industry Association of America. 2024 Year-End U.S. Latin Music Revenue Report. RIAA, 2025.
- [6] Spotify. Listeners Worldwide Crown Bad Bunny Global Top Artist for the Fourth Time. Spotify Newsroom, 3 Dec. 2025.
- [7] BBC News. Spotify Wrapped: Bad Bunny Tops Most-Streamed Artists Globally. BBC, 2022.
- [8] M. J. Salganik, P. S. Dodds, and D. J. Watts. Experimental study of inequality and unpredictability in an artificial cultural market. *Science*, 311(5762):854–856, 2006.
- [9] University of Texas at Austin. `rwd-billboard-data`. GitHub, accessed 1 June 2026.
- [10] University of Texas at Austin. `hot-100-current.csv`. `rwd-billboard-data`, GitHub, accessed 1 June 2026.